

From Reporting to Accountability: A Low-Tech Participatory Platform for Evaluating Government Climate Response

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Abstract

Climate change interventions often rely on expert-generated models and top-down planning processes that disregard local vulnerabilities and community knowledge. Participatory sensing low-tech digital platforms have been seen to bridge community experiential knowledge and institutional decision making in contexts such as the use of low-tech displays in public spaces to obtain actionable data on user experience. However, low-tech digital platforms rarely trigger government accountability which results in platforms that propagate power asymmetries and suppression of community voices. We examine how low-tech platforms evaluate institutional performance while legitimizing community experiential knowledge. Using MWAMKO as a case study, data was collected by community researchers to develop three modules. Module 1 provides a mechanism to document and report climate risks. Module 2 provides visualization and interpretation of processed data. From this co-creation process, a strong desire emerged to hold the government accountable. This birthed Module 3, developed to track reported cases and trigger government responsiveness.

CCS Concepts

• Human computer interaction (HCI); • Collaborative and social computing systems and tools; • ;

Keywords

Power asymmetries, participatory sensing, co-produced data, frugal HCI, community- centered design

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1 INTRODUCTION

Informal settlements in Nairobi continue to face climate-related risks such as flooding, extreme heat, and recurring disruptions to essential services that erode the safety and dignity of residents. Government responses to these risks remain slow and often harmful. Flood events are routinely followed by abrupt evictions carried out without resettlement plans, displacing households already living in precarious conditions. These actions are frequently justified as environmental protection measures, yet they occur despite advance early-warning forecasts from the Kenya Meteorological Department. The result is a persistent disconnect between available predictive knowledge and the timely, humane interventions these communities urgently require. In practice, climate risk in informal settlements arises not only from environmental exposure but also from opaque governance structures that leave residents without clear channels to communicate with authorities or track institutional action.

This governance gap is rooted in deeper structural conditions, including the absence of localized, community-specific data, fragmented institutional coordination, and limited mechanisms for monitoring climate impacts over time. Residents have no direct or consistent communication channels with agencies responsible for housing, infrastructure, and service delivery, leaving them unable to articulate needs to the very institutions tasked with protecting them. Climate risk in these environments is therefore socially produced through marginalization, information opacity, and persistent institutional inaction. While initiatives such as the Resilient Planet Initiative and DARAJA enable hazard mapping and weather alerts through SMS and social media, these tools largely rely on one-way information flows. In Mathare, residents reported that official reporting channels were unclear, inaccessible, and ineffective, with little or no follow-up even when reports were successfully submitted. This research argues that data collection and early warnings alone are insufficient, therefore systems are needed that make institutional responses visible and contestable.

Mathare Warning and Mitigation Knowledge Outreach (MWAMKO) offers a frugal, participatory, low-tech civic technological response to these systemic failures. Designed for resource-constrained settings, MWAMKO reimagines how knowledge and accountability move between communities and institutions by shifting residents from passive recipients of aid to active co-producers of evidence. Everyday observations such as flood depths, blocked drainage points, damaged footpaths, and

unsafe structures are transformed into structured, shareable data. Through accessible reporting tools, geolocation, and visualization, MWAMKO converts transient climate events into traceable records that communities can use to engage authorities and demand accountability, laying the groundwork for more responsive and inclusive climate governance

2 RELATED WORK

2.1 Civic Tech, Reporting Platforms and their Limits

Civic tech is defined as the use of appropriate digital technologies and methodologies in connecting government and citizens in public participation, transparency and accountability and in delivering public services [4]. In East Africa, a number of platforms focusing on climate risk information dissemination have been piloted majorly in informal settlements. RPI represents a comprehensive regional effort, combining community digital hubs, data portals, and mobile applications [6]. Participatory workshops with residents in Mukuru, Mathare and Kibera highlighted the need for offline USSD functionality to include households without smartphone access [7]. DARAJA exemplifies community-embedded early warning systems operating across Kenya and Tanzania, translating forecasts into accessible messages delivered through trusted intermediaries such as school clubs and women's groups [7]. Serving nearly one million users, the platform has significantly increased household preparedness. Complementary tools such as CLIMARK and Ushahidi support localized alerts and crowdsourced hazard mapping [8]. Across these initiatives, recurring design principles include reliance on frugal technologies, multi-channel communication strategies, and integration with existing social networks. However, most platforms stop short of enabling transparency in institutional response or accountability.

2.2 Critical & Participatory HCI in the Global South

The low-tech movement prioritizes user empowerment and technical minimalism but faces adoption challenges related to desirability, user burden and experience. Low-tech platforms are often perceived as less effective than conventional technologies, with their design dictated by the communities they intend to serve [1]. Critics argue that such approaches may lack scalability [2, 5]. Critical participatory HCI scholarship in the Global South foregrounds power symmetry, representation, and the co-production of knowledge, emphasizing lived experience and community agency. MWAMKO engages this literature by operationalizing low-tech principles pragmatically rather than ideologically, positioning participation, visibility, and accountability as central design objectives.

3 CONTEXT: CLIMATE RISK AND GOVERNANCE IN MATHARE

East African climate risk platforms operationalize low-tech principles out of necessity rather than ideology. Initiatives such as RPI and DARAJA demonstrate how participatory approaches function under infrastructural constraint, including intermittent electricity and limited smartphone access [3, 6]. In Mathare, climate risk is

shaped by both environmental exposure and governance failures. Despite early-warning forecasts, residents experience abrupt evictions, inadequate service responses, and limited participation in resilience planning. Fragmented institutional coordination and opaque decision-making processes reinforce socially produced vulnerability. This context underscores the need for platforms that go beyond information dissemination to enable accountability and community agency in climate governance.

4 METHODOLOGY

MWAMKO Resilience Mtaani was developed under a case study research design within Mathare informal settlement, Kenya between February to October 2025. The action research participatory design methodology employed was rooted in community-based research principles. Initial scoping involved interviews with Mathare residents (n=110) and interviews with community leaders (n=20) to identify data needs, communication patterns, and technological constraints.

The workshops focused on smartphone-based data collection, map interpretation, and evidence-based advocacy. Participants later served as peer trainers, supporting broader adoption. Module 1 and 2 of the platform launched publicly in August 2025 following a pilot phase. Iterative refinement was informed by monthly community meetings and in-platform feedback mechanisms. Planned improvements included simplified reporting categories, enhanced offline functionality, and improved media documentation quality.

Community feedback highlighted the need for customizable hazard categories, expanded language options, simplified interfaces, and stronger offline functionality, reflecting local conditions and connectivity constraints. Usage patterns indicated strong demand for accessible digital resilience tools, suggesting potential for expansion into similar informal settlement contexts.

However, residents identified three prerequisites for scaling: robust data privacy protection to prevent punitive use of reports, affordable access through zero-rated or low-cost data, and transparent communication on data governance. These findings emphasize that technical replicability alone is insufficient. MWAMKO's future success depends on addressing data sovereignty, cost barriers, and institutional responsiveness while building trust through transparent governance frameworks.

5 THE ARCHITECTURE OF MWAMKO PLATFORM

From the co-design process, three core goals were derived including; the platform should work on low-end phones and tolerate intermittent connectivity (frugal tech), the platform should be accessible to users with limited literacy or technology experience, and it should be able to generate outputs (maps, reports, visualizations) that both community members and government officials can understand. In practice, this meant providing both a web-based app and possibilities for plugging in of alternative channels such as SMS/USSD, designing a multilingual interface with intuitive icons, and structuring data for easy interpretation by any stakeholder.

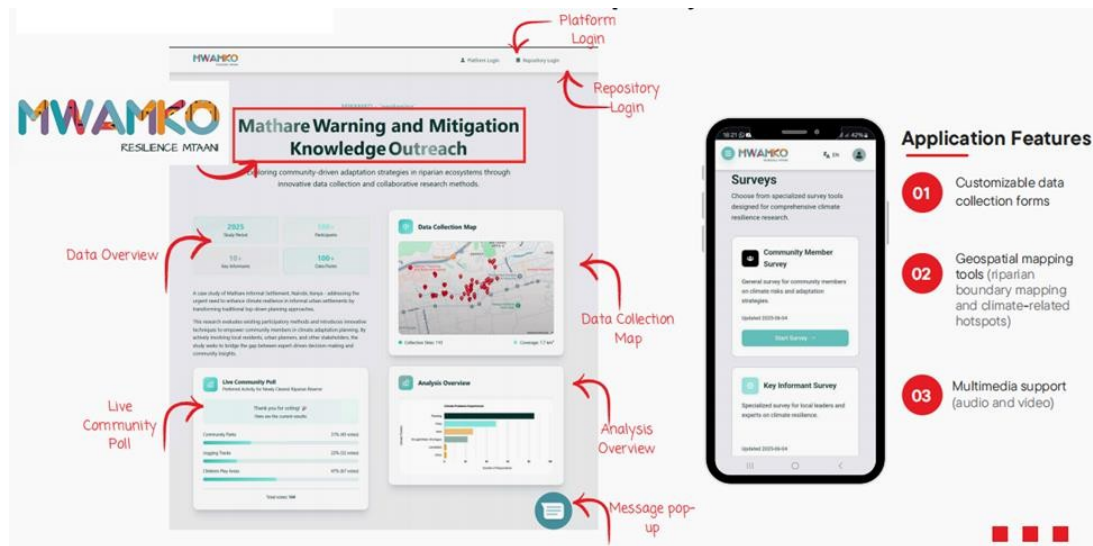


Figure 1: MWAMKO User Interface, a Co-created Prototype, Image by Author. [Public domain], via Resilience Mtaani. (<https://resiliencemtaani.co.ke/>)

5.1 System Architecture

MWAMKO is a socio-technical system integrating three core modules: community data collection, data visualization and analysis, and community-centered monitoring and reporting into a continuous cycle of evidence generation, interpretation, communication, and accountability. The first module provides structured mechanisms for residents to document climate risks such as flooding events, drainage failures, structural damage, displacement, and household vulnerability using geospatial and multimedia tools. Each submission is geotagged and enriched with photographs, videos, or audio testimonies, positioning residents as co-producers of risk information. The second module transforms community-generated data into spatial intelligence through interactive dashboards, layered maps, and automated reports. These outputs visualize hazard hotspots, flooding frequencies, and vulnerability profiles, supporting evidence-based planning by both community members and institutional actors.

The third module seeks to establish mechanisms for real-time reporting and institutional feedback. Reports will be assigned case IDs and publicly visible status updates, enabling residents to track responses and follow up on unresolved issues. The platform will allow end-to-end communication between the government and the persons reporting to create a continuous feedback loop on the efficiency of service delivery.

The envisioned evolution of MWAMKO focuses on strengthening its third module as a mechanism for translating community-generated evidence into actionable institutional response. This module will enable residents to report immediate climate and environmental hazards such as flooding, infrastructural deficits, and delayed government interventions directly through the platform. Each report will be assigned a unique case identifier, creating a publicly visible tracking system accessible to both communities and responsible authorities.

Government departments, utility providers, and humanitarian organizations are expected to acknowledge reports, post progress updates, and document completed actions, transforming reporting from a one-directional complaint process into an accountable feedback loop.

5.2 Why Low-Tech is a Feature, Not a Constraint

MWAMKO adopts a frugal, low-tech approach to align with Mathare's infrastructural realities. The platform functions on basic mobile phones, can integrate SMS-based interaction with minimal updates, and does not require constant internet connectivity. This design maximizes inclusion and enables participation across diverse community groups. Findings demonstrate that accessible tools can democratize environmental data and enhance responsiveness without undermining effectiveness. Simplified, minimalist interfaces and community SMS alerts further support timely hazard communication. In this context, low-tech design emerges as an enabling feature rather than a limitation.

6 RESULTS AND FINDINGS

Community-generated data identified flood-prone areas that closely correlated with GIS-based flood models, validating local knowledge as reliable evidence. Residents documented April–June as peak flood months, with 2024 identified as one of the most severe flood periods. The data enabled categorization of primary riparian risk zones and secondary low-lying areas prone to flash flooding, refining official risk mapping. Residents consistently described official reporting mechanisms as unclear and ineffective. During the eight-month pilot, hazard reports were noted by community researchers without tracked follow-up. County officials acknowledged receipt of reports but confirmed the absence of formal processing workflows, reinforcing perceptions of institutional inaction. Community members emphasized the need for a public-facing system that displays

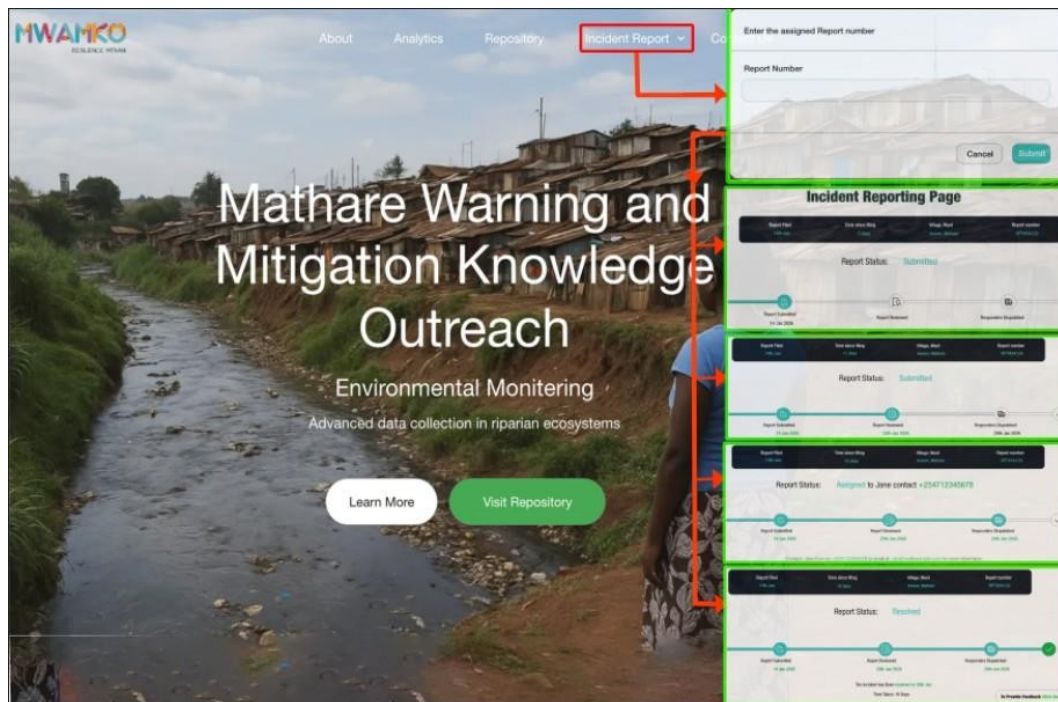


Figure 2: Module 3 Incident Reporting and Tracking, Image by Author. [Public domain], via Resilience Mtaani. (<https://resiliencemtaani.co.ke/>)

Figure 3: Citizen Incident Reporting Interface. Image by Author. [Public domain], via Resilience Mtaani. (<https://resiliencemtaani.co.ke/>)

reported issues and institutional responses. Time-stamped reports, visible acknowledgments, and response timelines were identified as critical features for restoring trust and perceived legitimacy. Following this, Module 3 development features took this feedback into account. Stakeholder mapping revealed reliance on community-based organizations during flood events, with limited presence of formal government agencies. Platform visualizations were widely appreciated for translating complex data into accessible formats, informing the design of the accountability module.

7 CONTRIBUTIONS TO CHI

This work contributes to HCI scholarship on participatory design by demonstrating how radical co-creation with marginalized communities produces accountability mechanisms that expert-driven systems consistently overlook. MWAMKO's three-module architecture emerged directly from residents' expressed needs, with the accountability tracking system developed only after communities articulated frustration with institutional unresponsiveness during the co-design process.

The platform illustrates how low-tech design can trigger government accountability through unique case IDs, publicly visible status tracking, and end-to-end communication channels that transform climate reporting from one-directional data extraction into a feedback loop making institutional action publicly legible. MWAMKO reconfigures power relations in climate governance, repositioning informal settlement residents from passive aid recipients to legitimate knowledge producers whose experiential evidence gains epistemic authority equivalent to expert models. This shifts control to the citizens, creating substantive channels for marginalized voices to contest technocratic climate interventions and demand government response rooted in lived experience.

8 CONCLUSION

MWAMKO demonstrates that frugal, participatory, and socially embedded technologies can support meaningful climate adaptation in informal settlements without reliance on sophisticated infrastructure. Through iterative co-design with residents in Mathare, the

platform positions communities as legitimate knowledge holders and surfaces critical requirements such as offline USSD functionality that expert-driven systems often overlook. The multi-channel architecture combining simple mobile and web low-tech early warning systems and trained resident monitors shows that effectiveness depends more on social embeddedness and trusted intermediaries than technological sophistication. Conceptually, this work redefines appropriate technology as tools tailored to local contexts and community strengths, challenging the idea that development requires complex technology. For policy and practice, MWAMKO illustrates that infrastructure deficits should be treated as design parameters rather than barriers. The central challenge is to design digital, analogue, or hybrid systems that enhance community agency rather than technological dependency. MWAMKO provides both a practical template and a conceptual foundation for such approaches in climate adaptation and beyond.

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